

Quiz 2 — answers and marking notes

1. (a) $\varphi_B(t) = (1 - p) + pe^{it} = 1 + p(e^{it} - 1)$.

(b) $\varphi_{N_n}(t) = \left[1 + \frac{\lambda}{n}(e^{it} - 1)\right]^n \rightarrow \exp(\lambda(e^{it} - 1))$, the characteristic function of $\text{Poisson}(\lambda)$.

(c) **Independence**, which is what gives $\varphi_{N_n} = \varphi_B^n$.

Marking. Part (c) is the one that matters and the one most likely to be skipped; the whole point of the week is that this step is where independence is spent. Accept “the $B_{n,i}$ are i.i.d.” but probe whether the student knows that *identical* distribution only lets them write one factor n times — it is independence that lets them multiply at all. Do not require the $(1 + z/n)^n \rightarrow e^z$ lemma to be quoted by name.

2. (a) $\mathbb{E}W = \sum_i \mathbb{P}(\pi(i) = i) = n \cdot \frac{1}{n} = 1$.

(b) **Independence of the indicators.** The I_i are dependent: $\mathbb{P}(I_1 = I_2 = 1) = \frac{1}{n(n-1)}$, whereas $\mathbb{P}(I_1 = 1)\mathbb{P}(I_2 = 1) = \frac{1}{n^2}$. Le Cam’s hypothesis is not satisfied, so the theorem says nothing here.

(c) A theorem whose hypotheses fail proves nothing, whatever the truth of its conclusion. The bound has to be obtained another way — in Homework 4, by a coupling.

Marking. This is the question that separates students who memorised a formula from students who read its hypotheses. Full credit in (b) requires naming *independence* specifically, not “the I_i aren’t nice” or “ W isn’t a sum of Bernoullis” — it is a sum of Bernoullis, which is exactly the trap. In (c), the sentence must locate the flaw in the *argument*, not in the conclusion; “the bound is loose” misses the point and earns nothing.

3. (a) $\mathbb{E}[W' - W \mid X_1, \dots, X_n] = -\frac{1}{\sqrt{n}} \mathbb{E}[X_I - X'_I \mid X_1, \dots, X_n] = -\frac{1}{\sqrt{n}} \left(\frac{1}{n} \sum_i X_i - 0 \right) = -\frac{W}{n}$.

(b) $a = 1/n$, by the tower property.

(c) Because the bound of Homework 4 needs only $\mathbb{E}[W' - W \mid W]$ and $\mathbb{E}[(W' - W)^2 \mid W]$ — conditional moments of a coupling, which can be computed without ever factorising $\mathbb{E}[X_i f(W^{(i)})]$ into $\mathbb{E}[X_i] \mathbb{E}[f(W^{(i)})]$.

Marking. In (a) the two things to check are that $\mathbb{E}[X'_I \mid X_1, \dots, X_n] = 0$ (the fresh copy is independent of everything conditioned on) and that averaging over I produces $\frac{1}{n} \sum_i X_i$. A student who writes $\mathbb{E}[X_I \mid X_1, \dots, X_n] = X_I$ has not seen that I is random and should lose (a) but may still earn (b) and (c). Accept any (c) that identifies the payoff as “only conditional moments of the pair are needed”.

Sources: Q1 from Homework 3, Problems 3 and 5 (core); Q2 from Homework 4, Problem 10 (core); Q3 from Homework 4, Problem 5 (core). Question types rotate against Quiz 1, which used COMPUTE, COMPUTE, WHICH HYPOTHESIS FAILS.